SEPARATION PROPERTIES OF J, THE STABILIZER, AND THE IDENTITY SECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma lemma-groupoid-stabilizer (0235) with schemes-lemma-separated-permanence (01KU). cited at proof lines 4376-4382. G -> U as base change of j; separatedness descends.
- Lemma schemes-lemma-section-immersion (01KT). cited at proof lines 4385-4389. Sections of separated morphisms are closed immersions.
- Lemma lemma-diagram-diagonal (02YH) with schemes lemmas 01JY and 01K5. cited at proof lines 4392-4395. The diagonal of R is a base change of e.

Source reference. Original lemma 0235: sources/stacks-chow/tags.txt, source lines 2730–2730. Cited at proof lines 4376-4380 to see G -> U is the base change of R -> U x_S U by the diagonal; load-bearing for (a) => (b).

Source reference. Original lemma 01KU: sources/stacks-chow/tags.txt, source lines 2063–2063. Cited at proof lines 4380-4382 (with 0235) for (a) => (b); load-bearing.

Source reference. Original lemma 01KT: sources/stacks-chow/tags.txt, source lines 2062–2062. Cited at proof lines 4385-4389 for (b) => (c); load-bearing.

Source reference. Original lemma 02YH: sources/stacks-chow/tags.txt, source lines 3876–3876. Cited at proof lines 4392-4394 for (c) => (a); load-bearing.

Source reference. Original lemma 01JY: sources/stacks-chow/tags.txt, source lines 2031–2031. Cited at proof lines 4392-4395 alongside 02YH; load-bearing.

Source reference. Original lemma 01K5: sources/stacks-chow/tags.txt, source lines 2038–2038. Cited at proof lines 4392-4395 for the quasi-separated case; load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid over S. Let $G \to U$ be the stabilizer group scheme.*

(1) *The following are equivalent*
   (a) *$j : R \to U \times_S U$ is separated,*
   (b) *$G \to U$ is separated, and*

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(c) $e : U \to G$ *is a closed immersion.*

(2) *The following are equivalent*

(a) $j : R \to U \times_S U$ *is quasi-separated,*

(b) $G \to U$ *is quasi-separated, and*

(c) $e : U \to G$ *is quasi-compact.*

Proof. The group scheme $G \to U$ is the base change of $R \to U \times_S U$ by the diagonal morphism $U \to U \times_S U$, see Lemma 0235. Hence if j is separated (resp. quasi-separated), then $G \to U$ is separated (resp. quasi-separated). (See Schemes, Lemma 01KU). Thus (a) $\Rightarrow$ (b) in both (1) and (2).

If $G \to U$ is separated (resp. quasi-separated), then the morphism $U \to G$, as a section of the structure morphism $G \to U$ is a closed immersion (resp. quasi-compact), see Schemes, Lemma 01KT. Thus (b) $\Rightarrow$ (a) in both (1) and (2).

By the result of Lemma 02YH (and Schemes, Lemmas 01JY and 01K5) we see that if e is a closed immersion (resp. quasi-compact) $\Delta_{R/U \times_S U}$ is a closed immersion (resp. quasi-compact). Thus (c) $\Rightarrow$ (a) in both (1) and (2). $\square$